

EM 519 - APPLIED ENGINEERING PROJECT MANAGEMENT

EMU COLLEGE OF TECHNOLOGY
School of Engineering Technology
Prerequisites: 509
Credit hours: 3 Semester Hours

Course objectives

At the completion of this course, the student should be able to:

- A. Explain practices and challenges in managing engineering projects and approaches to effective project management.
- B. Identify and plan the content for an engineering project using project management tools and techniques.
- C. Understand and describe the Project Management Institute's Body of Knowledge (PMBOK Guide).
- D. Apply project management principles, practices, and software for hands-on project planning, implementation, management, and analysis.
- E. Use project management tools for meetings and communications among project team members.
- F. Evaluate project team, team member, and team leader performance.

Course outline and schedule

Unit 1

Lecture:

Introduction to PMBOK
Project Life Cycle and Organization
Project Management Processes for a Project

Lab: Project initiation and basic planning.

Unit 2:

Lecture:

Project Integration Management
Project Scope Management

Lab: Project scope, pre-planning and execution.

Unit 3.

Lecture:

Project Time Management
Project Cost Management

Lab: Project plan, budget, update and execution.

Unit 4.

Lecture:

Project Quality Management

Lab: Project quality, update and execution.

Unit 5.Lecture:

Project Human Resource Management

Lab: Project resources, update, execution.

Unit 6.Lecture:

Project Communications Management

Lab: Project update, execution.

Unit 7.Lecture:

Project Risk Management

Project Procurement Management

Lab: Project risks, outsourcing, update, execution.

Unit 8.Lecture:

Final Exam, Post-test, and PMP/CAPM Practice Exam

Lab:

Project update, execution, and closure.

General policies

A. All reports should contain a title page, content page, a statement by the team if its a team work about the amount work done by each team members, a introduction explain the report and other details as you see fit for the report.

B. that Students are encouraged to discuss problems related to the class with each others and with their instructor(s) via telephone or email.

C. Academic Dishonesty: The policy of the School of Engineering Technology is that any student found to have engaged in any activity constituting academic dishonesty will receive an "E" for the course in which the activity occurred. This policy relates to all forms of work associated with the course requirements (tests, lab work, papers and other assignments). Academic dishonesty includes: cheating on quizzes and exams, copying, or taking credit for another's work, plagiarism, etc. Graduate students enrolled at EMU should be aware of academic dishonesty policies and consequences, and examples of what constitutes academic dishonesty, from EMU Academic Affairs, EMU Student Judicial Services, EMU Graduate School, and the School of Engineering Technology.

Our managerial approach regarding minor occurrences of academic dishonesty is to inform the involved student(s) about the problem, give a warning to the student(s) of the possible consequences, and share an example of the problem with the rest of the class, as a learning experience for the whole class. The consequences may be more severe (such as failure of the assignment or the entire course), and without a warning, for second and/or major occurrences of academic dishonesty.

D. Writing Style and Format: It is the policy of the School of Engineering Technology (SET) that students should follow the APA (American Psychological Association) technical writing style and format guidelines for papers submitted in all classes. Some professors will emphasize use of APA guidelines more than others. It is recommended that you get a copy of the 5th edition of the APA Publication Manual as soon as possible, as you will use it throughout your degree. You should especially be familiar with traditional and electronic referencing guidelines for quoting, paraphrasing, and referencing information that you use, to

help avoid Academic Dishonesty problems, as mentioned above. See <http://www.apastyle.org/styletips.html> and <http://www.apastyle.org/elecref.html> for more specific information and examples.

E. Student and Exchange Visitor Information Service (SEVIS): For **students who hold an F or J visa**, to avoid being out-of-status, it is your responsibility to report within 10 days to the Office of International Students (OIS), King Hall, any change in name, current address of residence, academic status (full or part-time enrollment), date of program completion, program/major change, level change (undergraduate to graduate), change in funding source (employment or graduate assistant position), probation or disciplinary action leading to suspension due to conviction of a crime. Effective January 1, 2003, EMU has 21 days from the date of the reported event to inform the U. S. Government Department of Homeland Security, Bureau of Border and Transportation Security and the Bureau of Citizenship and Immigration Services SEVIS database (functions of the formerly known as Immigration and Naturalization Service are a part of two new offices effective March 1, 2003).

OIS enters the data into a real-time database and new documents or I-20s are then issued to you. Remember that dropping or withdrawing from course(s) may put you below full-time enrollment. Before a drop/withdrawal change is made that reduces enrollment below full-time status, you must receive permission from the OIS. If you plan to transfer to another university, this process must begin before the end of your last semester at EMU, contact the Office of International Students. Noncompliance could result in actions leading up to a fine, arrest, or deportation per the U.S. Government.

F. Incompletes (I) will be given only in accordance with EMU's guidelines, which is stated in Graduate Catalog 2005-2007, p.19, or <http://www.emich.edu/public/catalogs/2005-2007/grad/2005all.pdf>.

I (Incomplete) Grades

An I grade may be awarded when illness or other unavoidable extenuating circumstances prevent completion of a course, provided that academic performance for the completed portion of the course merits a grade of B or better. The student must provide documentable proof that these circumstances were unavoidable, such as a note from a doctor, etc.

G. Graduate Students in Engineering Management program are expected to demonstrate high level of professionalism. Students' professional traits, such as courtesy and respect, good time and priorities' management, communication and interpersonal skills, honesty and dependability may also be taken into account when issuing final grade for the course. Instructor reserves the right to change grade either to a higher one or to a lower one based on instructor's discretion.

H: Post all issues with the class/course in the discussion thread "Course Problem" under Course Home, so that either your course mates or the instructor can answer it

Participation guidelines and requirements

Here are some guidelines for making things go smoothly.

A. Access to Course.

Our course consists of 8 units, which are scheduled from Mondays through Sundays, beginning on Wednesday, January 5, 2011 and ending on Sunday, April 16, 2011. **Assignments will be due on Sunday nights, by 11:59 PM ET US unless otherwise specifically stated.** Some assignments for each unit may be due before the end of the unit, so please read assignment pages carefully. I will usually open access to each unit early, on Friday nights, so that you can get an early start working on it if you like; therefore, you will have an extra weekend available for working on most of the units. All completed units will remain open for review, until the end of our class.

B. Participation.

In online courses, students seek interaction with other students and with the instructor. Since there are no required face-to-face meetings, this interaction becomes very important to effective learning online. As such, good class participation is essential and will account for a significant part of your grade. [Plan to contribute to every discussion topic with a "quality" contribution -- i.e. a response that expresses clear thinking, and that is relevant to the discussion. This usually involves more than just "I agree."]

C. Online Conduct.

Realize that the University and your instructors cannot and will not be responsible for the content of any personal messages that are sent from one student directly to another student using the online email delivery system. It is expected that all students will adhere to accepted codes of ethical, personal, and civil conduct when conversing online, using email, or engaging in any real time online chat sessions. Failure to abide by such codes of conduct and etiquette may result in withdrawal from the course and a failing grade.

D. Email to Classmates.

Students are free to use the "email the class" feature on the course web site. Try to keep the messages sent to the whole class related to course content. Also, when emailing the class, generally it is best NOT to email phone numbers or addresses unless it is being emailed directly to a specific individual or the instructor. This helps maintain privacy.

E. Email to Instructor.

For EM 519 related submissions and communications, please send all email to spopkov@emich.edu. If you need to reach my supervising professor - Dr. Ahmed, his e-mail is mahmed6@emich.edu. Also, in the subject line of your email messages, include the: **initials of your full name**, **course number**, and the **subject of the message** (such as the name of an assignment that is attached to your email); for example, Subject: **XYZ519Design1**.

I can not be on call for your class 24 hours a day, but will be checking in as often as possible. I expect to check EM 519-related email before 7 PM on several days during the week. Due the amount of email I usually receive from students, as well as other work and travel commitments, I may not be able to respond to emails or confirm receipt of submitted assignments immediately, although I will try to respond to your questions and concerns within a day. So, plan accordingly. If you wait until the last minute to do an assignment, and then email a request for assistance with a problem, neither I nor your fellow students may be able to respond quickly enough to help you.

F. Email Attachments.

There are so many different word processing programs that email attachments can become a problem in online courses if there is no standard policy set for them. It is preferred that you send any paper that is an attachment to an email in MS Word. Formats that are compatible with MS Word include earlier Word versions, Microsoft Works, Word Perfect 5 or 6, and Rich Text Format are also acceptable. If necessary, you can compose the work in your non-compatible word processor and simply cut and paste it into an email message--although this may cause a problem if formatting is important. If using MS Project software, please submit your work in MS Project 2007 format. You may also send single or multiple files in .zip compressed format.

The file name of your attachment should be similar to the subject line of your email message, including the: **initials of your full name**, **course number**, and the **name of the attachment with its file type**; for example, the file name of an attached assignment would look like: **XYZ695Presentation1.pdf**.

G. Grammar, Spelling, Writing Style, and Format.

We all tend to become somewhat lax when posting responses to threaded discussions or when sending email messages. Grammar and spelling are important. I am not a fanatic about it, but I would appreciate your composing threaded discussion responses or email messages in a word processing program where you can check spelling, edit, and revise your writing. Then you can cut and paste the work into the threaded

discussion or email message. You can do the same for other coursework and thereby help with NOT tying up Internet Service Provider lines.

Good grammar, spelling, writing style, and APA format is required for formal papers, per SET Department guidelines.

H. Assignment Delivery and Technical Problems.

Assignments that are submitted directly via the EMU Online e-College course system (such as threaded discussion responses, quizzes, and exams) must be submitted before midnight on Sunday night of their due date (by Sunday 11:59 PM ET US), and **will not be accepted late**. We will reluctantly accept late assignments of other types (those types of assignments that are supposed to be emailed directly to us or submitted through Drop Box), but they will be penalized **20% per day late (Monday 12:00 AM to 11:59 PM ET US is the definition of one day late)**. Since you will have more than two or three weeks available for completing most of the work throughout most of the course, we expect you to get it done on time. Final projects and the final exam (and any other assignments due in our Unit 5 final week) will be due by Sunday 08/16 (midnight), and will not be accepted late. Because of the nature of the EMU Online threaded discussion and examination and administration system, threaded discussions and exams taken or submitted late can not and will not be accepted.

We encourage you to make hard copies of your assignments and of your communications with me. Computer crashes and disasters are bound to happen. Keep in mind that we do not view ISP problems as an excuse for turning work in late. At the first sign of problems with email, contact your ISP immediately. If your problems are not cleared up within 12 hours, send in the assignment in one of the following ways:

1. Fax it.
2. Overnight express hard copy and on disk.
3. Make arrangements over the phone.

Note: It is your responsibility to access the course on another computer in the event of hardware or software problems. This can be done with a friend's computer, or one at Kinko's or the Public Library. My guidance from the university is to not consider any excuse for missing assignments due to technical difficulties as legitimate.

I. Quizzes and Exams.

Most of the quizzes and exams in this course are timed, based on a easy standard of 120 seconds (two minutes) per question. The typical standard for professional-level exams, such as certification or licensure exams, is about 60 to 90 seconds per question. What makes it even easier for you, is that your quizzes and exams in our course are very focused on specific information from sections of specific books--whereas certification and licensure exams cover a very wide body of professional knowledge, covered by numerous books and other material.

Like those professional exams, your quizzes and exams in this class are open-book (of course)--and, like those professional exams (and regular classroom exams), you should prepare for taking your quizzes and exams in advance, by reading the material to be covered, taking notes, and applying other test-preparation studying techniques.

J. Personal Involvement and Time.

Students often have the misperception that taking a course online is easier since they don't have to physically go to a classroom. When taking an online course there are some things that become easier for students. For example, it becomes easier NOT to go to class. It also becomes easier to get behind. This medium gives students flexibility in attending class--but remember, while taking a course online may be more convenient, it is not necessarily easier!

Be aware of the general guidelines to professors for student work loads: 45 hours of work per credit hour per semester. This is a 3 credit class, so I expect, on average, that students will spend 135 hours of total work time in this course. Students with more graduate-level and online learning experience tend to be more efficient and require less time--some students will require a great deal more time to get their school work completed. Be prepared for the work involved, as it may require 10-15 (or more) hours of your time each week.

Remember, this is a course for the Engineering Management program. Engineering Managers are responsible for planning, organizing, staffing, leading, and controlling engineering projects and personnel. They must lead by example, and make sure that objectives are met and completed on time. Many of our EM students have very busy professional lives and have significant work and family responsibilities in addition to going to school--yet they do their school work well, and get it done on time (often early). If a student in our program can not organize themselves well enough to get their school work done on time, then they will certainly have a problem organizing and leading other engineers and engineering projects, and probably should not be in this program. As such, we will give little leeway when it comes to assignments that are not submitted on time--and will suggest to chronically late students that they seek a different degree and profession.

Grading policies

EVALUATION.

Course grades will be determined by the percentage of total points attained for various work assignments completed throughout the semester, based upon the following scale:

100 to 93% = A
92.99 to 90% = A-
89.99 to 87% = B+
86.99 to 83% = B
82.99 to 80% = B-
79.99 to 77% = C+
76.99 to 73% = C
72.99 to 70% = C-
(no D grades, based on Grad School requirements)
69.99 and below = E

Final course grades will be based as shown above, but may be subject to adjustment (up or down) based on the judgment of your professor.

ACTIVITIES.

The general format of this class will be through online guidance from the professor(s) and students' submission of individual-based and team-based lecture and lab assignments. The lack of face to face discussions will have to be compensated for with threaded discussions and written feedback.

In some classes, during your Engineering Management program, you may participate in Case Studies solutions and debating various issues. Read the Unit Debate-Case to learn how to analyze case study and participate in debate.